

# FRUITS AND SEEDS

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CHERRIES, TOMATOES, and pea pods are all fruits. A fruit is the part of a plant that contains and protects the seeds.

A fruit forms after a flower has been pollinated. First the petals wither and fall, then the part of the flower called the ovary swells. This becomes the fruit, containing one or several seeds, which are the plant's way of reproducing itself. Inside the fruit, the seeds are supplied with nutrients through tiny stalks connecting them to the fruit wall. As the seeds grow, the fruit ripens. Some fruits are sweet and juicy and may be edible; others are inedible, or even poisonous.

## How a fruit develops

Once a flower has been pollinated and fertilization has taken place, its ovary becomes known as a fruit. This fruit and the tiny seeds within begin to develop and grow. Gradually the fruit enlarges, and as it matures, its shape, colour, and texture also change. When a juicy, edible fruit such as this melon ripens, its flesh becomes very sweet and succulent.

The flower is brightly coloured and attracts insects that will pollinate it.

After pollination, the ovary starts to swell.

The flower is no longer needed, so it shrivels up and dies.

The fruit begins to form.

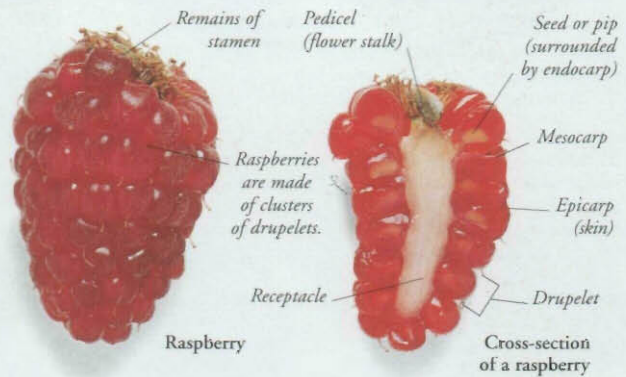
The fruit grows larger as it ripens.

The fruit has ripened and contains hundreds of seeds deep inside.

A melon is a kind of berry.

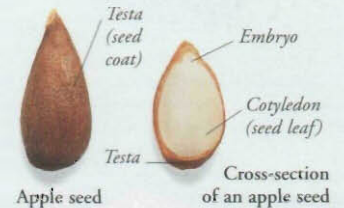
## Parts of a fruit

In some fruits, the fruit wall, or pericarp, has three distinct layers – an outer epicarp, a middle mesocarp, and a hard, inner endocarp. These layers are easy to see in fleshy fruits, such as plums, but in other fruits the layers are not so clear. The fleshy part of an apple, for example, is actually formed from the receptacle – the swollen tip of the flower stem.



## Parts of a seed

All seeds contain a tiny embryo and seed leaves called cotyledons, which are full of stored food. These are all enclosed in an outer seed coat, called a testa. The embryo has a minute root called a radicle and a tiny stem called a plumule. When the seed germinates, the food store provides nourishment for the tiny seedling.



## Seed dispersal

Plants need to spread their seeds to increase their chance of survival. Seeds are dispersed by wind, water, and animals. In some plants, parts of the fruit wall or flowerhead also help to spread the seeds. As the fruit dries, the fruit wall splits open and the seeds are scattered.

### Dispersal by burial

Seeds are a valuable source of food for mammals and birds. Squirrels and other rodents bury acorns and other nuts, then forget to dig them up. These grow into plants where they were left.



Squirrel burying nuts

### Bird dispersal

Brightly coloured orange and red berries attract birds, which like to feed on them. The birds swallow the berries whole but digest only the fleshy part. The seeds pass out, unharmed, in the bird's droppings.

### Redwing



### Water dispersal

Some fruits and seeds float. Their fruit wall contains oil droplets or air to make them buoyant. Coconut palm fruits float in the sea until they are washed up on a beach.



Coconut growing on a beach

### Wind dispersal

Light fruits and seeds are spread by the wind. The seeds of a columbine are scattered when the breeze shakes the seed head. Maple tree seeds have papery wings to carry them on the wind.



Columbine seed head

### Animal dispersal

Some seeds are encased inside hooked fruits that can easily become trapped in the fur of mammals. They are carried along by the animals and drop off later in another place.



Bison with seeds trapped in fur

## Types of fruit

Simple fruits have a single ovary; compound fruits have more than one. When ripe, some fruits remain succulent; others become woody and hard, or dry and papery, such as larkspur. False fruits develop from other flower parts in addition to the ovary.

### Berries

Berries have a combined mesocarp and endocarp layer. They often have many seeds that each have a tough seed coat, or testa.

### Grapes



### False fruits

In most false fruits, the receptacle swells to enclose the true fruit. Tiny true fruits may also be fixed to the surface of the receptacle.

### Pear



### Drupe

Drupe, or stone fruits, have a thick, fleshy mesocarp and a woody endocarp – the stone. Fruits such as raspberries are made up of many tiny drupelets.

### Plum



### Dry fruits

Dry fruits often have lids or seams that open to release the ripe seeds. This capsule has round openings called pores.

### Love-in-a-mist capsule



FIND OUT  
MORE

BIRDS

FLOWERS

FOOD

MAMMALS

PLANTS

PLANTS,  
ANATOMY

PLANTS,  
REPRODUCTION

PLANT  
USES

RATS AND  
OTHER RODENTS